Conditional Statement: (day-2)

Check whether a triangle:

Get 3 numbers from the user , compare the numbers and print Equilateral or Isosceles or Scalene

- If all number are same - Equilateral
- If any two numbers are same - Isosceles
- If all the numbers are different - Scalene

Hint : check the upper/lower case for the output

Input Format

Integer

Constraints

1 < n < 100

Output Format

Equilateral/Isosceles/Scalene

Sample Input 0

2 2 2

Sample Output 0

Equilateral

Explanation 0

all the numbers are same

Sample Input 1

4 5 4

Sample Output 1

Isosceles

```
import java.io.*;

import java.util.*;

import java.text.*;

import java.math.*;

import java.util.regex.*;
```

```
public class Solution {
   public static void main(String args[] ) throws Exception {
      Scanner sc = new Scanner(System.in);
      int a = sc.nextInt();
      int b = sc.nextInt();
      int c = sc.nextInt();
      if(a==b && b==c){
         System.out.println("Equilateral");
      }
      else if(a==b || b==c || a==c){
         System.out.println("Isosceles");
      }
      else{
         System.out.println("Scalene");
      }
   }
}
```

Electricity Bill Calculator 12:

Electricity is charged as follows:

First 100 units: ₹5 per unit

Next 100 units: ₹7 per unit

Beyond 200 units: ₹10 per unit

Given total units U, calculate and print the total bill. Use if–else if statements.

Input Format

An integer U.

Constraints

Nill

Output Format

Total bill as an integer.

Sample Input 0

50

Sample Output 0

250

Sample Input 1

125

Sample Output 1

675

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int units= sc.nextInt();
        int bill=0;
        if(units<=100){
            bill=units*5;
        }else if(units<=200){
            bill=100*5+(units-100)*7;
        }else{
            bill=100*5+100*7+(units-200)*10;
        }
        System.out.println(bill);
    }
```

}

Simple Calculator 50:

Accept two integers a, b and a character op representing an operator (+, -, *, /, %). Perform the corresponding operation and print the result. Use a switch statement. If division by zero occurs, print Error.

Input Format

a b arthimatic operator

Constraints

Nill

Output Format

Result of the operation or Error.

Sample Input 0

10

5

/

Sample Output 0

2

Sample Input 1

10

90

+

Sample Output 1

100

```
import java.io.*;

import java.util.Scanner;

import java.text.*;

import java.math.*;

import java.util.regex.*;


public class Solution {
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int a = sc.nextInt();
    int b = sc.nextInt();
    String op = sc.next();
    switch(op){
      case "+":
        System.out.println(a+b);
        break;

      case "-":
        System.out.println(a-b);
        break;

      case "*":
        System.out.println(a*b);
        break;

      case "/":
        System.out.println(a/b);
        break;

      case "%":
        System.out.println(a%b);
        break;

      default:
        System.out.println("Error");
```

```
    }
   }
}
```

Conditional Statement -1:

Write a C program to check the number is either divisible by 3 or divisible by 5 and display the corresponding message

Input Format

Accept an integer as an input

Constraints

NIL

Output Format

Hi/Hello/HiHello

Sample Input 0

1500

Sample Output 0

HiHello

Sample Input 1

93

Sample Output 1

Hi

Sample Input 2

25

Sample Output 2

Hello

```
import java.io.*;
import java.util.Scanner;
import java.text.*;
import java.math.*;
import java.util.regex.*;
```

```
public class Solution {


    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt();
        if(a%3==0 && a%5==0){
            System.out.println("HiHello");
        }else if (a%3==0 && a%5 !=0){
            System.out.println("Hi");
        }else{
            System.out.println("Hello");
        }
    }
}
```

Bus Fare Based on Route and Time:

City buses charge different fares:

Route	Morning	Evening
A	25	30
B	20	25
C	15	20

Input Format

Route letter (A/B/C)

Time in 24-hr format integer (e.g., 9, 18).

Constraints

Nil

Output Format

Integer fare.

Sample Input 0

B

19

Sample Output 0

25

Sample Input 1

A

6

Sample Output 1

25

Sample Input 2

C

18

Sample Output 2

20

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        char c = sc.next().charAt(0);
        int t = sc.nextInt();
        switch(c){
            case 'A':
```

```
            if(t>=1 && t<=12)
               System.out.println("25");
            else
               System.out.println("30");
            break;
            case 'B':
            if(t>=1 && t<=12)
               System.out.println("20");
            else
               System.out.println("25");
            break;
            case 'C':
            if(t>=1 && t<=12)
               System.out.println("15");
            else
               System.out.println("20");
            break;
      }
   }
}
```

Character Category 1:

You are given a single character C.

Your task is to identify and print the category of the given character based on the following conditions:

If the character is an uppercase vowel (A, E, I, O, U) → print Upper Vowel

If the character is an uppercase consonant → print Upper Consonant

If the character is a lowercase vowel (a, e, i, o, u) → print Lower Vowel

If the character is a lowercase consonant → print Lower Consonant

If the character is a digit (0–9) → print Digit

Otherwise → print Special Character

Input Format

A single character C

Constraints

A single character C

Output Format

Print the category of the given character.

Sample Input 0

A

Sample Output 0

Upper Vowel

Sample Input 1

b

Sample Output 1

Lower Consonant

Sample Input 2

7

Sample Output 2

Digit

Sample Input 3

@

Sample Output 3

Special Character

```
import java.io.*;
import java.util.*;

public class Solution {
   public static void main(String[] args) {

      Scanner sc = new Scanner(System.in);
```

```
        char c = sc.next().charAt(0);

        if (c >= 'a' && c <= 'z') {

            if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') {
                System.out.println("Lower Vowel");
            } else {
                System.out.println("Lower Consonant");
            }

        } else if (c >= 'A' && c <= 'Z') {

            if (c == 'A' || c == 'E' || c == 'I' || c == 'O' || c == 'U') {
                System.out.println("Upper Vowel");
            } else {
                System.out.println("Upper Consonant");
            }

        } else if (c >= '0' && c <= '9') {

            System.out.println("Digit");

        } else {

            System.out.println("Special Character");
        }
    }
}
```

Ticket Price Calculator 1:

A theatre charges different prices based on age and day of the week.

Children (<12) : ₹100 on weekdays, ₹80 on weekends

Adults (12–59): ₹150 on weekdays, ₹120 on weekends

Seniors (60+): ₹90 on weekdays, ₹70 on weekends

Write a program to calculate the ticket price.

Input Format

First line: Integer age

Second line: String day (Monday–Sunday)

Constraints

Nil

Output Format

Single integer – the ticket price.

Sample Input 0

10

sunday

Sample Output 0

80

Sample Input 1

25

monday

Sample Output 1

150

```
import java.io.*;

import java.util.*;

import java.text.*;

import java.math.*;

import java.util.regex.*;
```

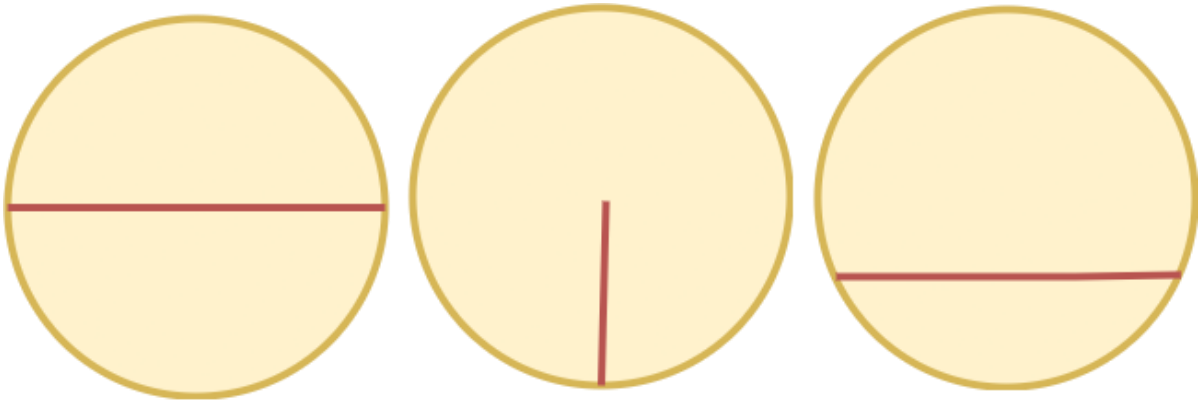
```
public class Solution {

    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        String str=sc.next();
        if(n<12){
            switch(str){
                case ("sunday"):
                    System.out.println(80);
                    break;
                case ("saturday"):
                    System.out.println(80);
                    break;
                default:
                    System.out.println(100);
                    break;
            }
        }else if(n>=12 && n<60){
            switch(str){
                case ("sunday"):
                    System.out.println(120);
                    break;
                case ("saturday"):
                    System.out.println(120);
                    break;
                default:
                    System.out.println(150);
                    break;
```

```
        }
        }else{
            switch(str){
                case ("sunday"):
                    System.out.println(70);
                    break;
                case ("saturday"):
                    System.out.println(70);
                    break;
                default:
                    System.out.println(90);
                    break;
        }
        }
    }
}
```

Minimum Cuts to Divide a Circle:

A cut that is represented by a straight line that touches two points on the edge of the circle and passes through its center, or A cut that is represented by a straight line that touches one point on the edge of the circle and its center. Some valid and invalid cuts are shown in the figures below.



Valid cut Valid cut Invalid cut

Input Format

integer

Constraints

1 <= n <= 100

Output Format

integer

Sample Input 0

4

Sample Output 0

2

Sample Input 1

3

Sample Output 1

3

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        if(n==1){
            System.out.println(0);
        return;
        }
        if(n%2==0){
```

```
            System.out.println(n/2);
        }else{
            System.out.println(n);
        }
    }
}
```